

Assurance-Lattice Receipts for Fail-Closed Ethereum Virtual Machine Inference Credit: A Reproducible Development Study

Anonymous Development Draft

Authorship and affiliations pending accountable-human determination

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DEVELOPMENT DRAFT — NOT SUBMISSION READY. Independent novelty, protocol, provenance, reproduction, authorship, repository, license, fee, and accountable-human approval gates remain open.

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ABSTRACT

Blockchain-mediated artificial-intelligence inference needs more than evidence that computation occurred: credit activation can also depend on semantic disposition, authority, provenance, freshness, evidence origin, claim scope, and exact request–response bindings. We study a typed assurance-lattice receipt whose Ethereum Virtual Machine credit gate uses a non-compensating meet rule: every required condition must pass, while failure or unknown status rejects activation with a deterministic reason code. The development prototype comprises a separately transcribed Python oracle, a Python reference kernel, and a Solidity 0.8.24 gate. We exhaustively executed all 729 states in a six-factor, three-level model and seven prespecified binding mutations. The local development run observed 0/728 false activations, 1/1 activation of the all-pass state, 729/729 base decision-and-reason matches, 7/7 mutation rejections, and 736/736 Python-generated-vector versus Solidity matches. A second same-owner local process reproduced byte-identical core result artifacts, and a later digest-pinned hermetic producer replay passed the Python and offline Foundry paths. These findings establish only development-level conformance to the frozen finite model. Novelty remains unresolved, the protocol was not independently verified before execution, the attack set is not exhaustive, no gas comparator was frozen, and no external or field validation was performed.

Keywords: blockchain assurance, Ethereum Virtual Machine, verifiable artificial-intelligence inference, fail-closed authorization, provenance, reproducible software evaluation

1 INTRODUCTION

Optimistic and deterministic approaches already provide mechanisms for verifiable blockchain-based machine-learning inference, including fraud-proof-style verification and public re-execution (Conway et al., 2024; Ribeiro Alves et al., 2026). Signed tool receipts can distinguish direct evidence from inference (Basu, 2026), while an attestation study shows that omitted freshness checks can permit replay despite otherwise valid evidence (Sokolov, 2026). Governance-first architectures further show how deterministic execution gates, evidence chains, and explicit assumptions can be evaluated as systems properties (Calboreanu, 2026).

These predecessors defeat any claim that receipts, deterministic gates, optimistic inference verification, provenance, or freshness are individually new. The narrower hypothesis examined here is whether their decision-relevant dispositions can remain typed and independently failing through an Ethereum

34 Virtual Machine receipt-to-credit transition, rather than collapsing into a compensating confidence score.
35 This remains a novelty hypothesis pending an independently owned search challenge and feature-level
36 reconciliation with the strongest prior art.

37 The development contribution is a formal fail-closed rule, a minimal Python/Solidity artifact, an
38 exhaustive finite-state evaluation, and a provenance-preserving bundle of positive, negative, and
39 inconclusive evidence. The study does not claim a new inference-proof primitive, semantic truth, production
40 consensus safety, field impact, or publication readiness.

2 METHOD

41 2.1 System and receipt rule

42 The assurance-lattice receipt contains six assurance dimensions: execution, semantic, authority,
43 provenance, freshness, and origin/scope. Each takes PASS, FAIL, or UNKNOWN. Seven exact predicates
44 cover unused-receipt status, request and response hash bindings, authority-epoch liveness, allowed evidence
45 origin, claim-scope agreement, and type-tag validity. Eligibility is their conjunction (Figure 1).

46 2.2 Deterministic gate

47 Algorithm 1: Fail-closed assurance-lattice-receipt eligibility.

- 48 1. For dimensions in the frozen order execution, semantic, authority, provenance, freshness, and
49 origin/scope, return its typed FAIL or UNKNOWN reason at the first non-pass state.
- 50 2. Check, in order, that the receipt is unused; request and response hashes match; the authority epoch
51 is live; evidence origin is allowed; claim scope matches; and the type tag is valid. Return the
52 corresponding rejection reason at the first failed predicate.
- 53 3. Return ELIGIBLE only if no rejection occurred.

54 The frozen order makes diagnostics deterministic. Consequently, reason-frequency counts reflect
55 precedence, not empirical risk or relative dimension importance.

56 2.3 Design and analysis

57 The base design is the complete Cartesian product $3^6 = 729$, a finite population rather than a statistical
58 sample. The all-pass tuple is the only eligible base state. Seven additional fixtures introduce one exact-
59 binding defect at a time. A separately transcribed oracle generates expected vectors outside the Python
60 reference kernel; those vectors are compiled into the Solidity test. The oracle, kernel, and Solidity
61 implementation nevertheless share the same frozen specification and producer, so common-mode error
62 remains possible. Missing or malformed rows count as failures, and no rows are excluded or imputed.
63 P-values and observed power are inapplicable.

64 2.4 Implementation and reproducibility

65 The development run used Python 3.14.6, Foundry 1.5.1-stable, and Solidity 0.8.24 on macOS 15.1
66 arm64. The runner captured argument vectors, return codes, logs, and SHA-256 values without network
67 access. A second same-owner process replayed the workflow with a fixed protocol timestamp (Figure 2).
68 A candidate harness now copies frozen inputs into a fresh temporary directory, verifies source hashes,
69 re-executes the evaluation, and compares exact result hashes. A subsequent producer-controlled digest-
70 pinned Linux amd64 container replayed the Python evaluation, exact Solidity 0.8.24 compilation, both
71 Foundry parity tests, figures, and manuscript build with network disabled, read-only root and case mounts,
72 dropped capabilities, and no new privileges. Both producer-side workflows passed, but neither is external
73 reproduction or independent verification.

74 2.5 Project information required by the article type

75 **Project link and archival identifier:** not yet approved or created; this blocks submission. **Operating**
76 **system:** development execution on macOS 15.1 arm64; producer-controlled hermetic replay on Linux
77 amd64 under Docker Desktop emulation. **Programming languages:** Python 3.14.6 and Solidity 0.8.24.

78 **Execution tooling:** Foundry 1.5.1-stable locally and digest-pinned Foundry 1.5.1-v1.5.1 in the hermetic
79 replay. **License and restrictions for non-academic use:** unresolved pending accountable-human source-
80 authority and licensing review. No public availability is promised by this draft.

81 2.6 Ethics and evidence boundary

82 The evaluation used synthetic state fixtures and locally authorized existing source material. It involved
83 no participants, personal data, live networks, or financial transactions. Institutional and venue-specific
84 determinations remain pending accountable-human review.

85 2.7 Artificial-intelligence assistance

86 OpenAI Codex assisted with local code generation, execution, analysis, visual-source preparation, and
87 drafting. Exact product model/version records and the final venue-compliant disclosure must be verified by
88 accountable human authors. Artificial intelligence is not listed as an author; all scientific claims, citations,
89 figures, and potential plagiarism require human verification before submission.

3 RESULTS

90 All 729 base states executed. The 728 ineligible states produced zero activations; the all-pass state activated.
91 Expected and observed base decisions and reasons agreed for 729/729 states. All seven binding mutations
92 rejected, and all 736 shared Python-generated vectors matched Solidity outputs (Table 2). Three Python
93 unit tests and two Foundry tests passed.

94 The same-owner replay reproduced identical SHA-256 values for the base CSV, mutation CSV, and
95 Python summary. Timing-bearing logs and run manifests differed as expected. The hermetic replay returned
96 zero for all eight declared commands, reproduced the frozen Python hashes, passed both offline Foundry
97 tests, and produced a venue-neutral PDF byte-identical to the earlier pinned TeX build. These checks
98 remain producer-owned. Figure 3 reports the first-reason distribution; its 486/162/54/18/6/2/1 pattern
99 follows the fixed precedence and is not an empirical risk ranking.

100 Several decisive outcomes remain negative or unresolved: independent protocol review did not precede
101 execution; no numeric gas comparator was frozen; malformed-type parity is partial; independent
102 reproduction and field validation were not performed; novelty is unresolved; and the prior proof-of-
103 inference semantic result remains NOT_SUPPORTED with a false-acceptance rate of 0.500 against a frozen
104 0.25 threshold. The prior negative is retained as context and not used as positive assurance-lattice-receipt
105 evidence.

106 3.1 Examples of use and limitations

107 The prototype demonstrates one bounded use: deterministically refusing local credit when any typed
108 assurance dimension is failed or unknown, or when one prespecified exact binding fails. It does
109 not demonstrate semantic correctness, universal malformed-input coverage, decentralized adversaries,
110 production consensus safety, privacy, trusted-execution-environment behavior, mainnet execution, user
111 outcomes, or field impact. The finite model and mutation set may be incomplete; neither producer replay is
112 independent replication; and the derived image is content-addressed locally but not publicly distributed.

4 DISCUSSION

113 Within the frozen model and seven named mutations, the Python and Solidity implementations obeyed the
114 prespecified fail-closed rule. This narrow conformance result does not establish that the model captures
115 every assurance dimension or attack, that semantic input is correct, or that local behavior transports to
116 production. Assurance-case research likewise cautions that successful test processes do not automatically
117 warrant inferences from tested to fielded behavior (Richard et al., 2026).

118 Novelty is especially uncertain. LATTICE overlaps deterministic governance gates and auditable evidence
119 chains; tool receipts overlap typed evidence sources; freshness work overlaps replay-resistant evidence
120 currency; and opML/EigenAI overlap blockchain inference verification and enforcement. The remaining
121 candidate difference is the specific typed, non-compensating Ethereum Virtual Machine receipt-to-credit

composition and its exhaustive development contract. Whether this is material or obvious requires independent scientific judgment.

4.1 Scalability and boundary conditions

The 3^6 base matrix is tractable because the prototype deliberately fixes six ternary dimensions. Exhaustive enumeration grows exponentially if new typed dimensions are added. The current result therefore supports a finite contract, not unrestricted assurance scaling. Foundry test-call gas is neither transaction gas nor production gas; no efficiency claim is made. A fair baseline and bounded gas protocol must be frozen before any scalability comparison.

4.2 Limitations and next falsification

The protocol and analysis plan were reviewed only within the producing artificial-intelligence-assisted workflow. The bounded literature search awaits an independent challenge, patent/standard refresh, and full-text predecessor reconciliation. A prospective unchanged rerun after independent protocol review is the cheapest next falsification step. External reproduction can then test whether the artifact and instructions suffice outside the producer's environment. Authorship, declarations, repository authority, licensing, venue fee, and submission approval remain open.

5 CONCLUSION

The prototype provides exact development evidence for one deliberately narrow claim: within the frozen finite model and seven named binding mutations, separately implemented Python and Solidity decision paths conformed to the same fail-closed rule. That result is a falsifiable software contract, but it is not evidence of semantic correctness, universal attack coverage, production safety, external generality, or material novelty. The minimum responsible publication path is therefore to preserve this claim ceiling while completing independent novelty and protocol review, a prospective unchanged rerun, and external reproduction.

CONFLICT OF INTEREST STATEMENT

The conflict-of-interest declaration is unresolved and must be completed and approved by all accountable authors before submission.

AUTHOR CONTRIBUTIONS

Authorship eligibility, order, affiliations, initials, and CRediT roles are unresolved. No author-contribution statement is approved in this development draft.

FUNDING

Funding and sponsor roles are unverified and must be completed by accountable authors before submission.

ACKNOWLEDGMENTS

OpenAI Codex assisted with local code generation, execution, analysis, visual-source preparation, and drafting. Accountable human authors must verify the tool name, version, model, source, affected content, citations, figures, and plagiarism status before approving the final acknowledgment.

DATA AVAILABILITY STATEMENT

Development code, generated state tables, logs, manifests, figure data, diagram source, and manuscript source are preserved locally in the research-system folder. No public repository, archival DOI or URI, license, or immutable release has been approved; public availability is therefore pending human decision, not promised.

SUPPLEMENTAL DATA

156 The minimum artifact bundle is prepared locally but is not authorized for public release. Its final public
157 contents, licenses, exclusions, and persistent identifier require accountable-human approval.

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TABLES

Table 1. Development conditions and analysis populations.

Condition	Definition	<i>n</i>	Excluded
Base	Complete six-dimension, three-level Cartesian product	729	0
Mutations	One failed binding with all six dimensions passing	7	Compound faults
Replay	Core artifacts compared across two same-owner processes	3	Timing logs

Table 2. Prespecified development outcomes. Intervals are not shown because the enumerated state space is a finite population rather than a sample.

Outcome	Numerator	Denominator	Development decision
False activations among ineligible states	0	728	Pass
Activation of all-pass state	1	1	Pass
Base decision and reason agreement	729	729	Pass
Mutation rejection	7	7	Pass
Python-generated vector/Solidity agreement	736	736	Pass

FIGURES

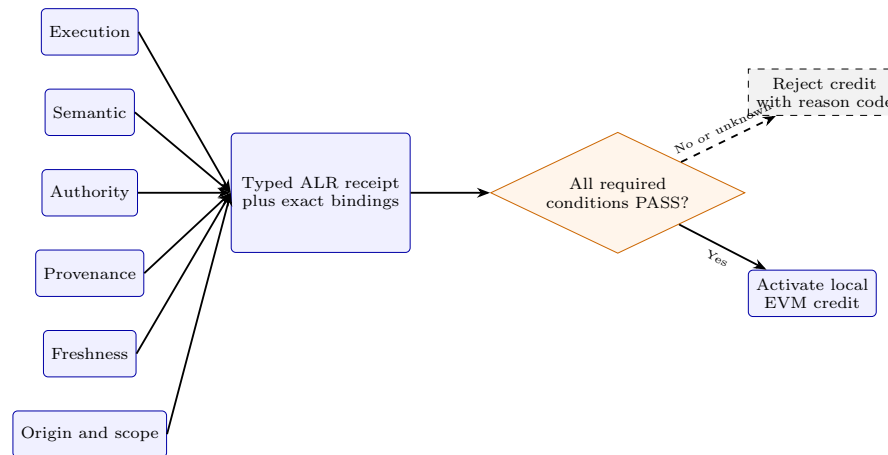


Figure 1. Assurance-lattice-receipt development architecture. Six typed assurance dimensions and seven exact-binding predicates feed a non-compensating gate. Failure or unknown status rejects credit with a deterministic reason; only all-pass input activates local Ethereum Virtual Machine credit. The diagram is architectural, not empirical evidence. Alt text: six assurance inputs enter a typed receipt and an all-pass decision diamond that branches to deterministic rejection or local credit activation.

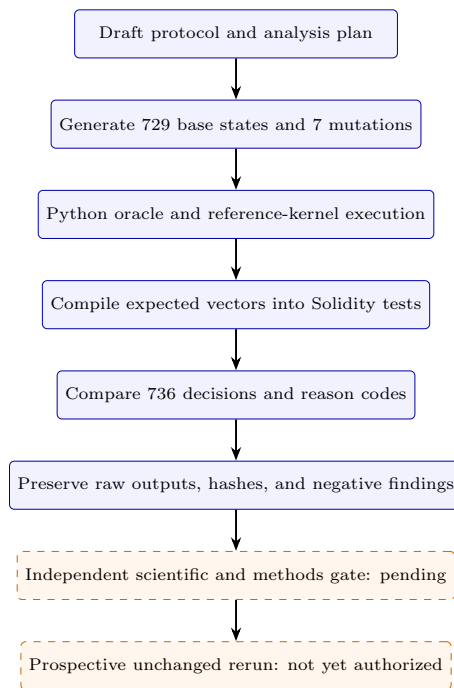


Figure 2. Development workflow and open gate. The current evidence stops before independent scientific and methods verification and a prospective unchanged rerun. Alt text: a vertical sequence runs from draft protocol through state generation, Python and Solidity comparison, and evidence preservation, ending at two dashed pending-review stages.

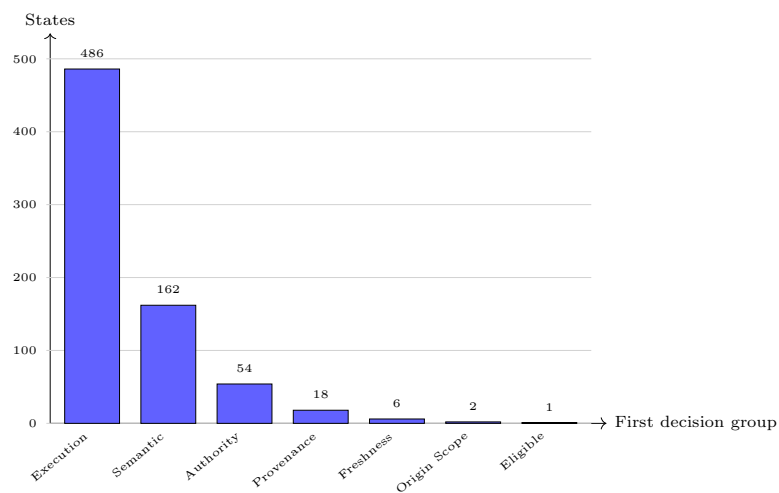


Figure 3. First-decision reason groups across the complete 729-state base matrix. Counts are deterministic products of Cartesian enumeration and the frozen precedence rule; they do not measure prevalence, severity, or assurance importance. Alt text: seven bars descend from 486 execution rejections to 162 semantic, 54 authority, 18 provenance, 6 freshness, 2 origin/scope, and 1 eligible state.